QSM I Assignment 1, Question 1

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Packages and Data Preparation

To make sure that all models (`m1`, `m_quad`, `m_comp`, and `m_full`) are estimated on the same sample size ($N = 68$) and prevent dimension mismatches during diagnostic testing, we filtered for complete observations.

```
# Load required packages
library(readxl)
library(car)
library(sandwich)
library(lmtest)

# Load dataset
cars_raw <- as.data.frame(read_excel("data_problemset1.xls", sheet = "Car Prices"))

# Rename variables clearly
names(cars_raw) <- c("CC", "HP", "Year", "Km", "Gas", "ABS", "ABD", "CD", "AW", "AC", "MP", "Price")

# Filter complete cases across all variables to standardize sample size (N = 68)
vars_all <- c("Price", "Km", "CC", "HP", "Year", "Gas", "ABS", "ABD", "CD", "AW", "AC", "MP")
cars_clean <- cars_raw[complete.cases(cars_raw[, vars_all]), ]
```

```
m_comp <- lm(Price ~ Km + HP + Year + Gas, data = cars_clean)
summary(m_comp)
```

```
##
## Call:
## lm(formula = Price ~ Km + HP + Year + Gas, data = cars_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6082.4 -1535.3  -383.3  1908.6 10859.7
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.364e+06  2.595e+05  -9.109 4.20e-13 ***
## Km          -3.719e-02  9.904e-03  -3.755 0.000380 ***
## HP           7.774e+01  7.370e+00  10.547 1.50e-15 ***
## Year         1.189e+03  1.297e+02   9.171 3.28e-13 ***
```

```
## Gas         -3.964e+03  9.568e+02  -4.143 0.000104 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3181 on 63 degrees of freedom
## Multiple R-squared:  0.8595, Adjusted R-squared:  0.8506
## F-statistic: 96.39 on 4 and 63 DF,  p-value: < 2.2e-16
```

```
## (f) OLS vs. Robust Standard Errors
```

(F) Comparing regular OLS SEs with Heteroskedasticity-consistent SEs for m_comp.

```
# Regular OLS standard errors
coeftest(m_comp)
```

```
##
## t test of coefficients:
##
##                Estimate  Std. Error t value  Pr(>|t|)
## (Intercept) -2.3635e+06  2.5948e+05 -9.1087 4.201e-13 ***
## Km          -3.7192e-02  9.9035e-03 -3.7555 0.0003800 ***
## HP           7.7737e+01  7.3703e+00 10.5473 1.503e-15 ***
## Year         1.1893e+03  1.2969e+02  9.1707 3.284e-13 ***
## Gas         -3.9644e+03  9.5682e+02 -4.1433 0.0001039 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
print(summary(m_comp), digits = 10)
```

```
##
## Call:
## lm(formula = Price ~ Km + HP + Year + Gas, data = cars_clean)
##
## Residuals:
##           Min            1Q        Median            3Q           Max
## -6082.3861289 -1535.2728787  -383.3322377  1908.5548221 10859.6940824
##
## Coefficients:
##                     Estimate       Std. Error  t value   Pr(>|t|)
## (Intercept) -2.363512787e+06  2.594790734e+05 -9.10868 4.2013e-13 ***
## Km          -3.719227002e-02  9.903520406e-03 -3.75546 0.00038002 ***
## HP           7.773706776e+01  7.370312109e+00 10.54732 1.5027e-15 ***
## Year         1.189327237e+03  1.296878344e+02  9.17069 3.2837e-13 ***
## Gas         -3.964358606e+03  9.568208256e+02 -4.14326 0.00010394 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3181.039 on 63 degrees of freedom
## Multiple R-squared:  0.8595449902,   Adjusted R-squared:  0.8506272118
## F-statistic: 96.38555159 on 4 and 63 DF,  p-value: < 2.220446e-16
```

```r
# Robust HC0 standard errors
coeftest(m_comp, vcov = vcovHC(m_comp, type = "HC0"))
```

```
##
## t test of coefficients:
##
##                Estimate  Std. Error t value  Pr(>|t|)
## (Intercept) -2.3635e+06  2.8142e+05 -8.3984 7.187e-12 ***
## Km          -3.7192e-02  1.0608e-02 -3.5062 0.0008435 ***
## HP           7.7737e+01  9.4326e+00  8.2413 1.351e-11 ***
## Year         1.1893e+03  1.4067e+02  8.4550 5.727e-12 ***
## Gas         -3.9644e+03  9.3931e+02 -4.2205 7.970e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
ct <- coeftest(m_comp, vcov = vcovHC(m_comp, type = "HC0"))
print(unclass(ct), digits = 10)
```

```
##                      Estimate      Std. Error      t value        Pr(>|t|)
## (Intercept) -2.363512787e+06 2.814245632e+05 -8.398388399 7.187148032e-12
## Km          -3.719227002e-02 1.060752660e-02 -3.506215108 8.435348945e-04
## HP           7.773706776e+01 9.432623915e+00  8.241298334 1.351132732e-11
## Year         1.189327237e+03 1.406659680e+02  8.454974957 5.726706781e-12
## Gas         -3.964358606e+03 9.393141501e+02 -4.220482152 7.970133121e-05
## attr(,"method")
## [1] "t test of coefficients"
## attr(,"df")
## [1] 63
## attr(,"nobs")
## [1] 68
## attr(,"logLik")
## 'log Lik.' -642.3086232 (df=6)
```

Comparison Table

Variable	Coefficient ($\hat{\beta}$)	Regular OLS SE	Regular p-val	Robust (HC0) SE	Robust p-val	Difference in SE
(Intercept)	-2,363,512.79	259,479.07	< 0.001	281,424.56	< 0.001	21,945.49
Km	-0.0372	0.0099	< 0.001	0.0106	< 0.001	0.0007
HP	77.74	7.37	< 0.001	9.43	< 0.001	2.06
Year	1,189.33	129.68	< 0.001	140.67	< 0.001	10.99
Gas	-3,964.36	956.82	< 0.001	939.31	< 0.001	-17.51

Findings

Coefficient estimates and R^2 (85.95%) remain identical under both methods. Accounting for heteroskedasticity means that we allow for the variance of the error term to differ across observations so the usual OLS formula misstates the true sampling variability of the estimates.

We can see that for km, HP and year that the standard error increased, this supports our test

Because larger SEs give smaller t-statistics, the robust p-values are higher. However, looking at the robust p-values, we can see that they still remain under 0.01, meaning we will reject the null hypothesis that the coefficients value is zero, with a significance of up to 1%.

(g) Executive Report for Used-Car Sales Management

To: General Manager, Used-Car Sales
From: Econometrics & Analytics Team
Date: October 2026
Subject: Vehicle Valuation & Future Market Strategy

Executive Summary

Our econometric regression analysis ($R^2 = 85.9\%$) indicates that while mileage steadily reduces vehicle values, baseline vehicle specifications—specifically **fuel type, engine power, and model age**—drive the main price premiums in the used car market.

Statistical Findings & Actionable Recommendations

1. **Prioritize Diesel Inventory Sourcing**
    - *Statistical Evidence:* Holding mileage, power, and age constant, we can be very confident gasoline cars sell for around **€3,964 less** than equivalent diesel models ($p < 0.001$).
    - *Action:* When buying stock, a diesel is the better purchase as long as it costs us less than about **€4,000** more than an equivalent petrol car, because we can expect to recover that premium when we sell. When pricing our own stock, diesel cars can carry an asking price around **€4,000** above comparable petrol cars.
2. **Market High-Horsepower Trims Prominently**
    - *Statistical Evidence:* Each additional unit of horsepower increases selling price by **€77.74** ($p < 0.001$).
    - *Action:* If cars have a higher HP, be sure to use this as a markup premium, as we have seen that customers consistently pay more for cars with higher horsepower.
3. **Value Model Age Over Low Mileage**
    - *Statistical Evidence:* Each year newer adds around **€1,189** to car price ($p < 0.001$), while each 1000 km takes off around **€37**. This means that one extra design year is therefore worth roughly as much as 32,000 fewer kilometres.
    - *Action:* When sourcing stock, prioritize newer model years even if they carry moderately higher mileage over older, low kilometres units. When pricing, don't discount newer cars heavily for high mileage, because buyers pay mainly for design year.
4. **Avoid Non-Essential Cosmetic Features**
    - *Statistical Evidence:* Optional features including CD players, alloy wheels, air conditioning, ABS, air bags and metallic paint **do not display statistically significant individual price premiums** ($p > 0.05$), either individually or when combined.

- *Action:* Do not allocate funding to retrofit accessories, as buyers do not pay a statistically significant premium for these additions.
